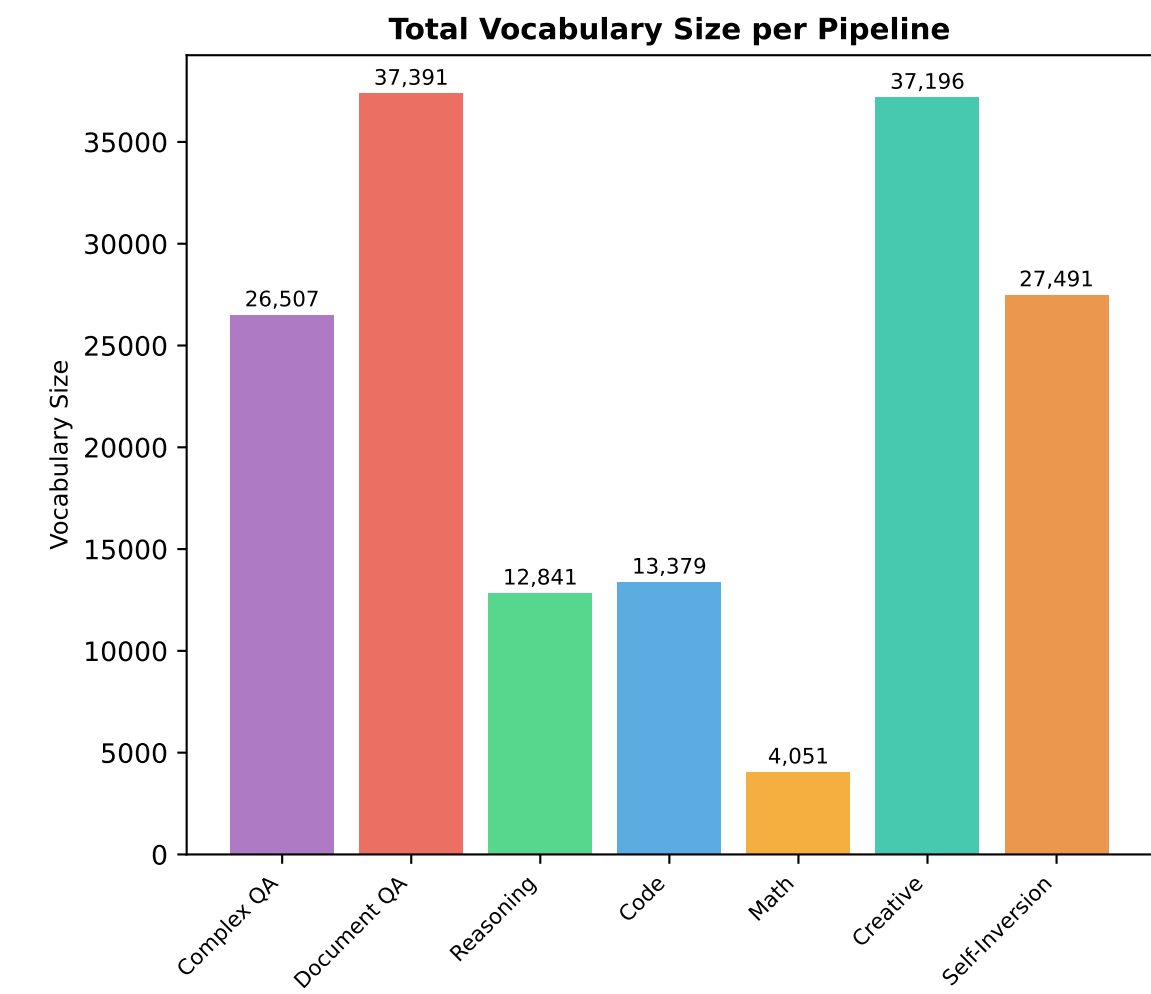
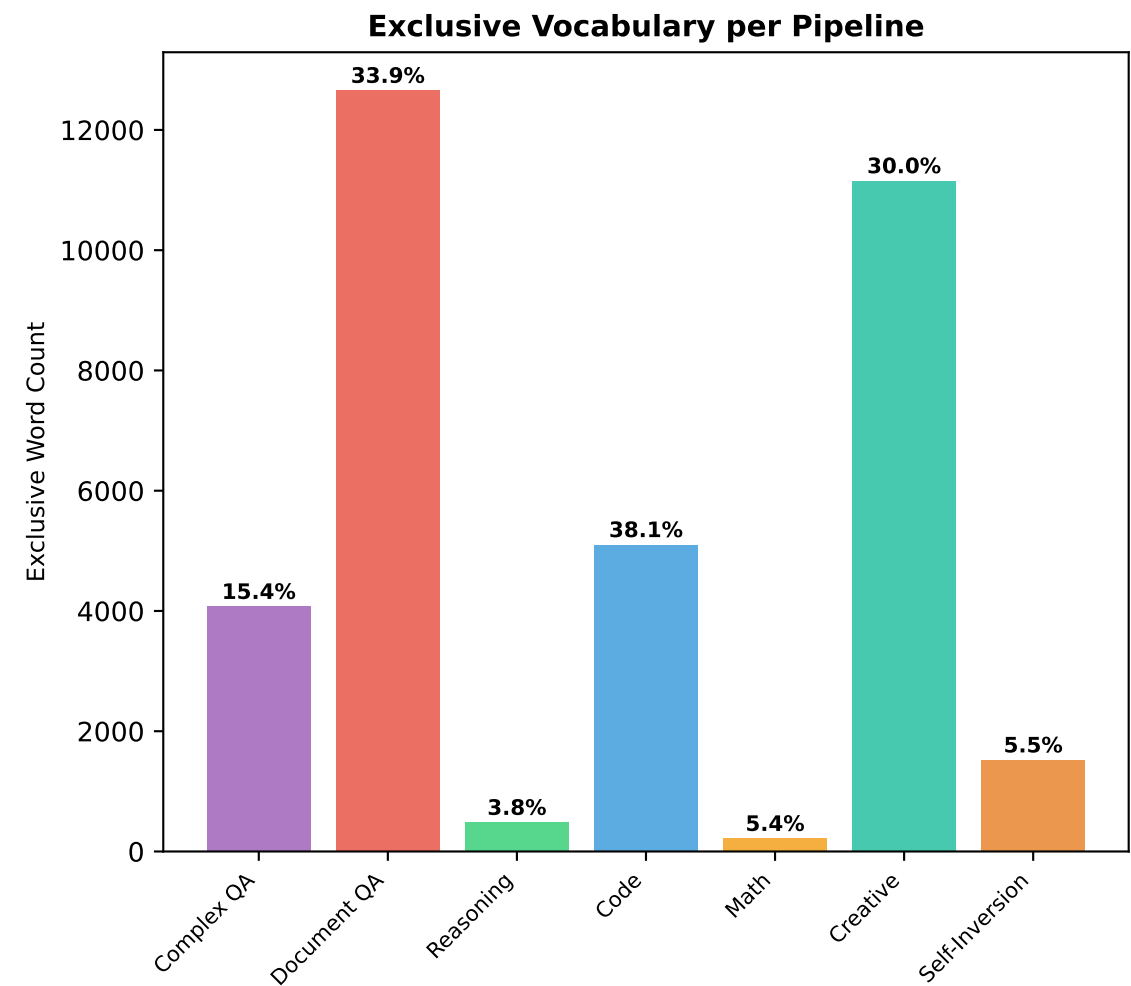
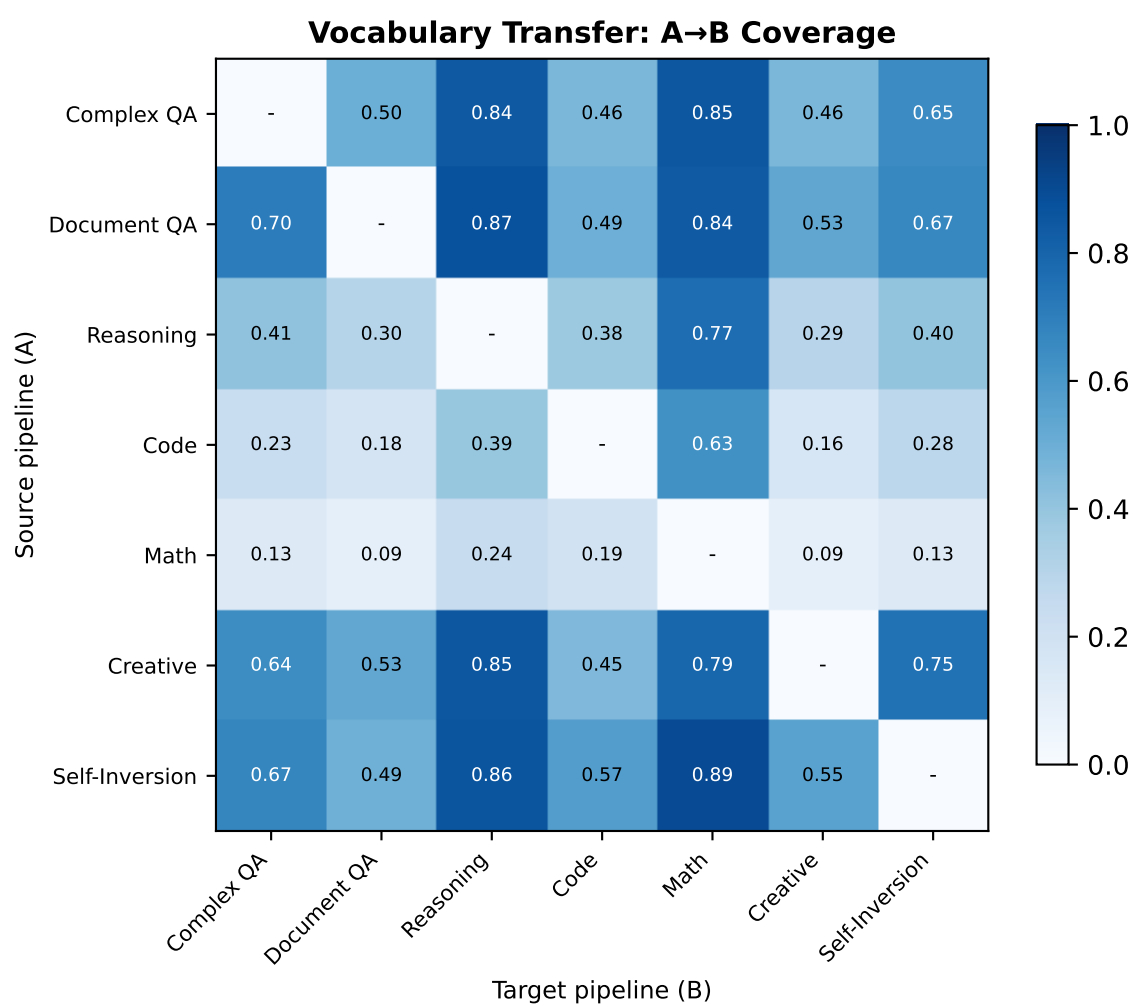
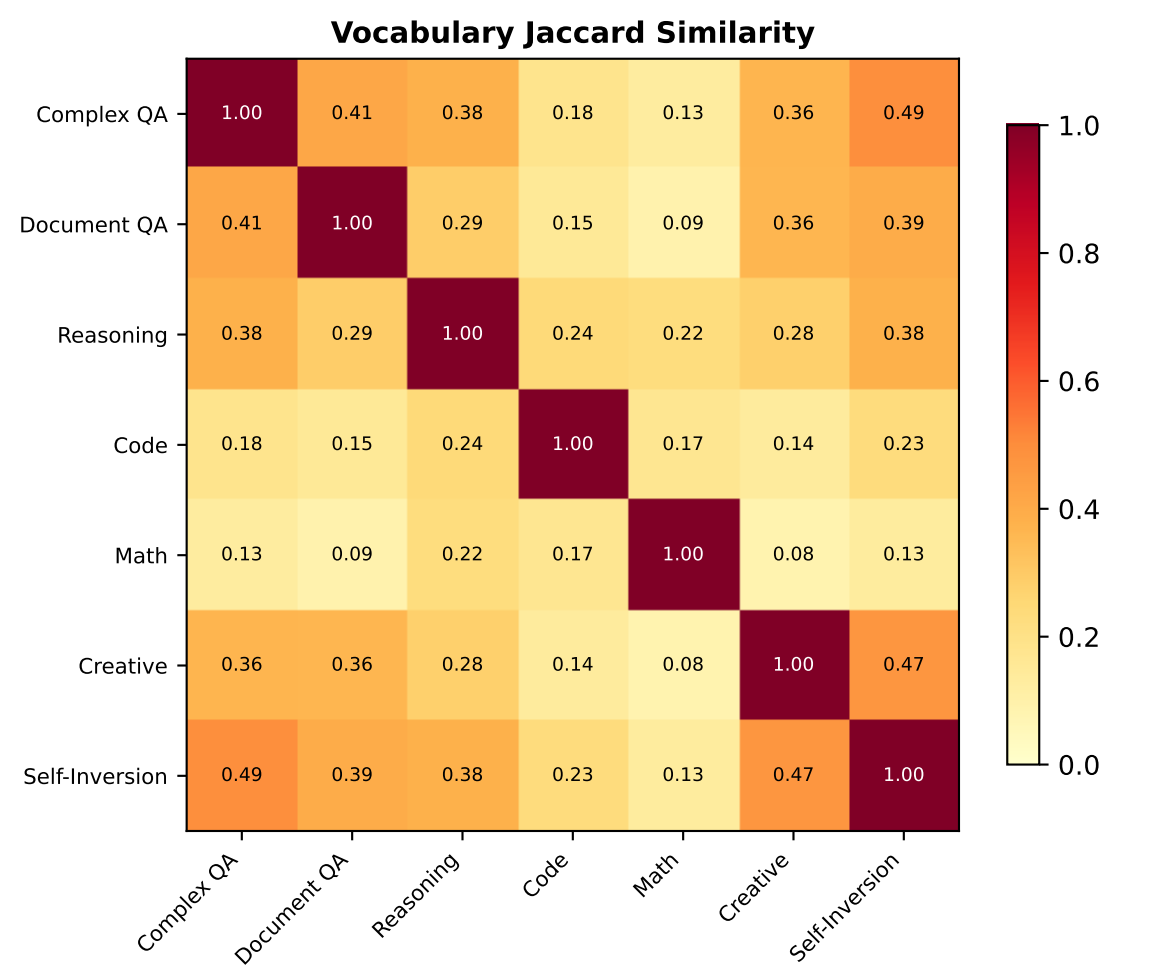
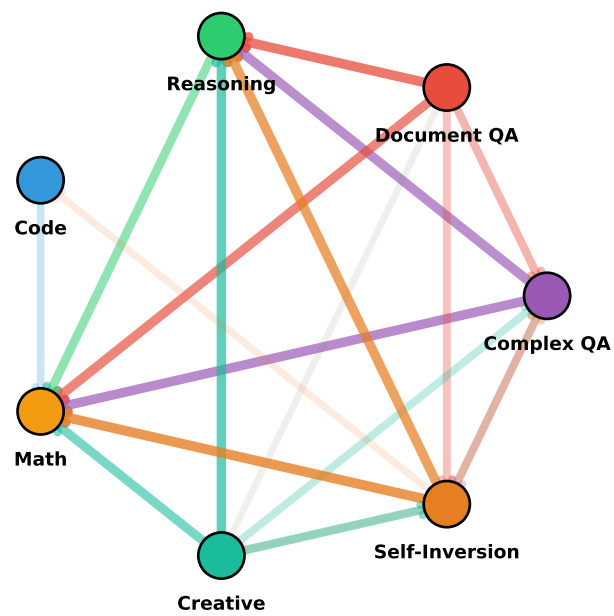


Fig 169: Cross-Pipeline Vocabulary Transfer Analysis



Vocabulary Transfer Network (>50% coverage)



Top TF-IDF Distinctive Words per Pipeline

Complex QA:
economic(0.0004), periodization(0.0004), carbon(0.0003), energy(0.0003), emissions(0.0003), bim(0.0003)

Document QA:
cultural(0.0005), emotional(0.0004), economic(0.0004), community(0.0003), legal(0.0003), engagement(0.0003)

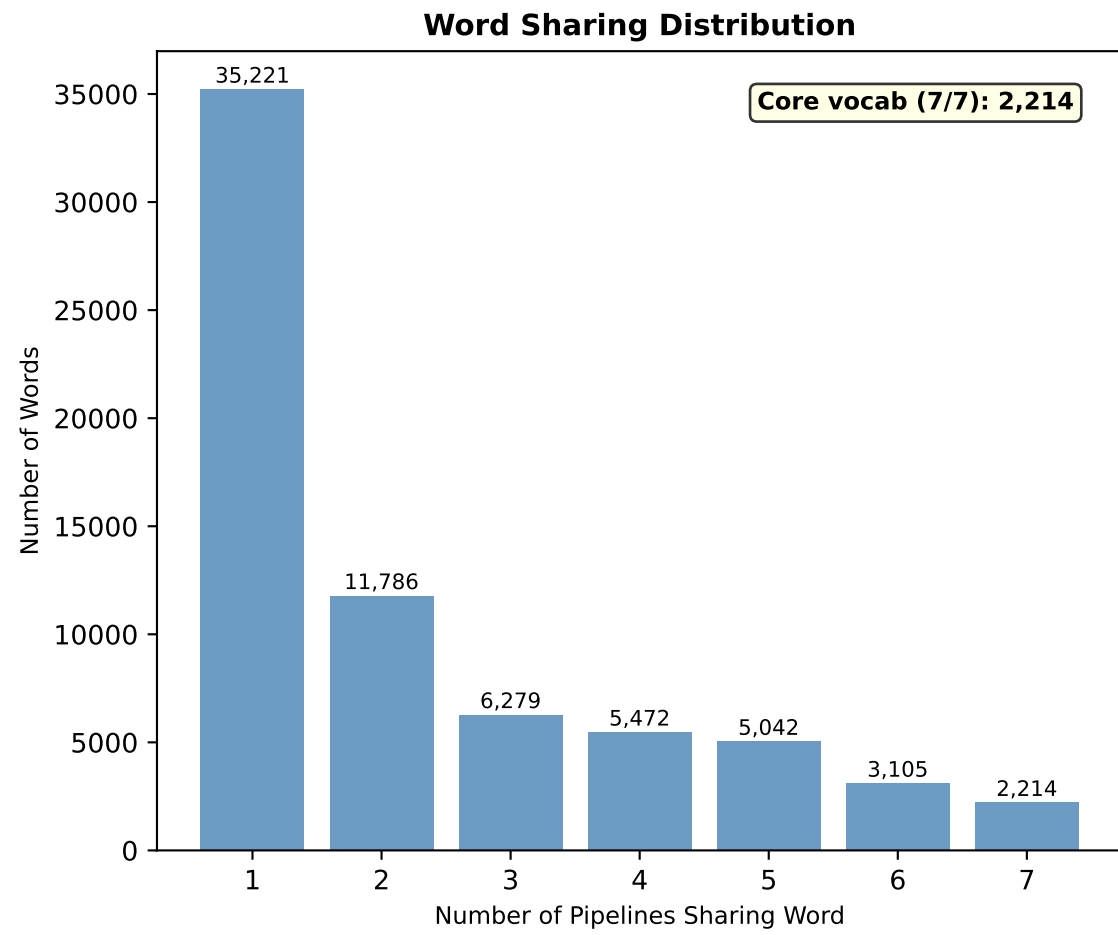
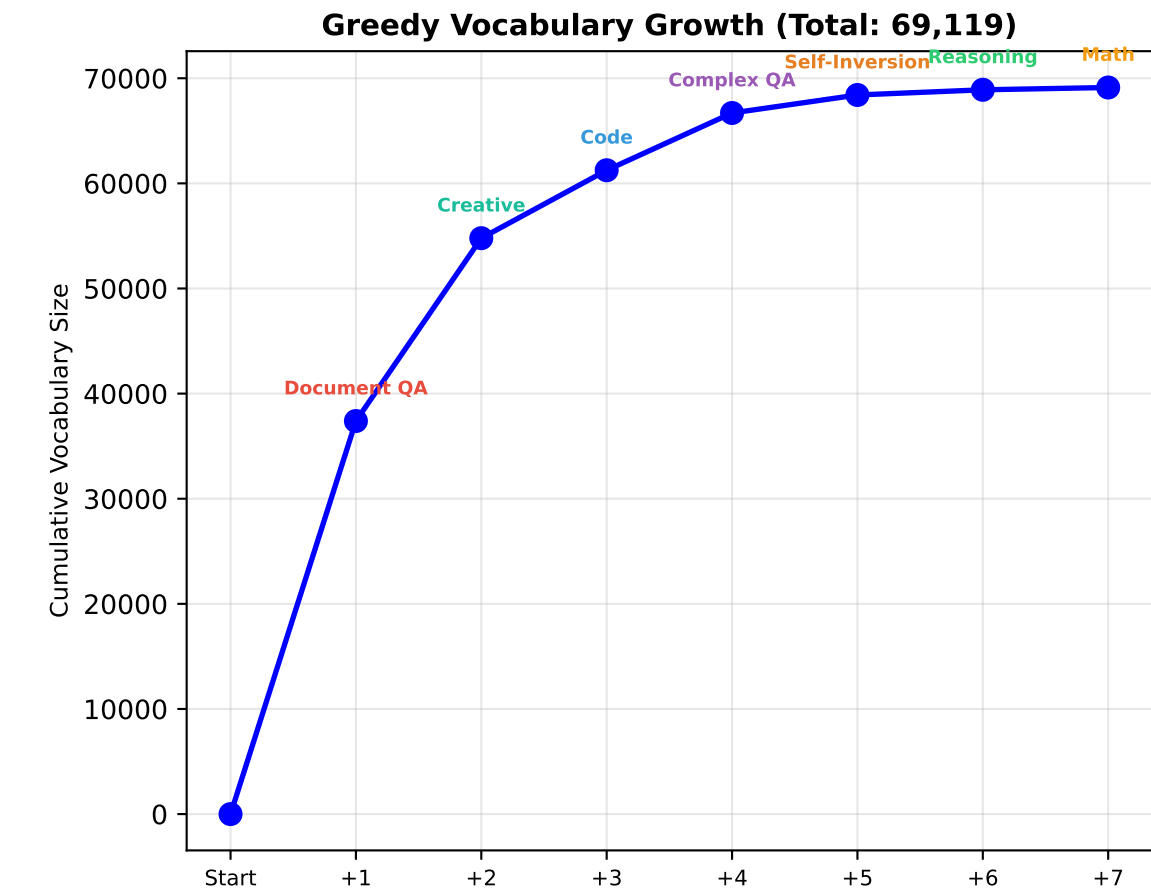
Reasoning:
barber(0.0013), premises(0.0008), shave(0.0007), casey(0.0006), disease(0.0005), shaves(0.0005)

Code:
const(0.0037), std(0.0030), func(0.0023), singleton(0.0018), println(0.0016), import(0.0014)

Math:
sylow(0.0036), pmatrix(0.0024), circ(0.0022), frac(0.0020), equiv(0.0014), vmatrix(0.0014)

Creative:
her(0.0018), elara(0.0009), she(0.0009), his(0.0007), dreams(0.0005), eyes(0.0005)

Self-Inversion:
const(0.0007), singleton(0.0007), std(0.0006), func(0.0006), her(0.0004), import(0.0003)



Vocabulary Transfer Summary

Total unique vocabulary: 69,119
Average Jaccard similarity: 0.265
Average transfer coverage: 0.505
Core vocabulary (shared by all): 2,214 (3.2%)
Most exclusive vocab: Code (38.1%)
Best vocabulary source: Document QA
Greedy order: Document QA -> Creative -> Code -> Complex QA -> Self-Inversion -> Reasoning -> Math

Key Findings:

- 2,214 words shared across all 7 pipelines
- Code/Math have most specialized vocabulary
- Complex QA/Document QA best vocabulary donors
- Self-Inversion bridges vocabulary gaps